PW AITS
ALL INDIA TEST SERIES

Dropper JEE Main

AJPM10/09 **Test- 09**

DURATION : 180 Minutes **DATE : 02/03/2025** **M.MARKS : 300**

Topics Covered

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Mathematics:	Full Syllabus

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 75 questions. The maximum marks are **300**.
4. There are three Sections in the question paper, Section I, II & III consisting of Section-I **(Physics)**, Section-II **(Chemistry)**, Section-III **(Mathematics)** and having **25 questions** in each part in which first **20** questions are of Objective Type and Last **5 questions** are integers type and all **25 questions are compulsory**.
5. There is only one correct response among 4 alternate choices provided for each objective type question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.
9. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ **Invigilator's Signature** _____

IMPORTANT CONSTANTS

Speed of light in free space,	:	$3.00 \times 10^{8}\ ms^{-1}$
Permeability of free space,	:	$4\pi \times 10^{-7}\ Hm^{-1}$
Permittivity of free space,	:	$8.85 \times 10^{-12}\ Fm^{-1}$
The Planck constant,	:	$6.63 \times 10^{-34}\ Js$
Rest mass of electron,	:	$9.1 \times 10^{-31}\ kg$
Rest mass of proton,	:	$1.67 \times 10^{-27}\ kg$
Molar gas constant,	:	$8.31\ JK^{-1}\ mol^{-1}$
The Avogadro constant,	:	$6.02 \times 10^{23}\ mol^{-1}$
The Boltzmann constant,	:	$1.38 \times 10^{-23}\ JK^{-1}$
Gravitational constant,	:	$6.67 \times 10^{-11}\ N\ m^2kg^{-2}$
Acceleration of free fall	:	$9.8\ ms^{-2}$
Rydberg Constant	:	$1.097 \times 10^{7}\ m^{-1}$
Atomic mass unit	:	$1.67 \times 10^{-27}\ kg$
Charge on proton	:	$1.6 \times 10^{-19}\ C$

IMPORTANT VALUES

$\sqrt{2}$ = 1.414

$\sqrt{3}$ = 1.732

$\sqrt{5}$ = 2.236

π = 3.142

e (Euler's constant) = 2.718

ln 10 = 2.303

$\log_{10}2$ = 0.3010

$\log_{10}3$= 0.4770

$\log_{10}7$ = 0.845

**** Use above values unless otherwise specified in a question.***

❑ ❑ ❑

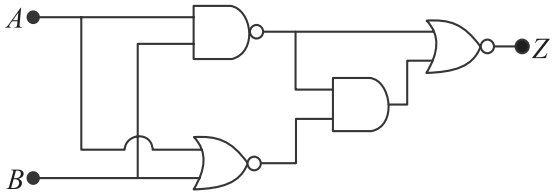
SECTION-I (PHYSICS)

Single Correct Type Questions

1. $\left(P+\frac{a}{V^2}\right)(V-b)=RT$ represents the equation of state of some gases, where P is the pressure, V is the volume, T is the temperature and a, b, R are the constants. The physical quantity, which has dimensional formula as that of $\frac{a}{b}$, will be

(1) Power (2) Surface Tension
(3) Work (4) Energy density

2. In the following, digital circuit, what will be the output at 'Z', when the input (A, B) are (1, 0), (0, 0) (1, 1), (0, 1)

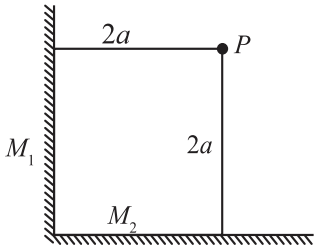


(1) 0, 1, 0, 0 (2) 1, 0, 1, 1
(3) 1, 1, 0, 1 (4) 0, 0, 1, 0

3. A helicopter rises from rest from the ground vertically upwards with a constant acceleration 2g. A food packet is dropped from the helicopter when it is at a height h. The magnitude of velocity of the packet through which it strikes on the ground [g is the acceleration due to gravity]

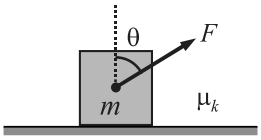
(1) $v=\sqrt{6\,hg}$ (2) $v=3.4\sqrt{hg}$
(3) $v=\sqrt{2hg}$ (4) $v=\sqrt{3hg}$

4. Two plane mirrors M_1 and M_2 are at right angle to each shown. A point source 'P' is placed at '$2a$' and '$2a$' meter away from M_1 and M_2 respectively. The shortest distance between the images thus formed is:



(1) $2\sqrt{2}a$ (2) $2a$
(3) $4\,a$ (4) Zero

5. A block of mass m slides along a floor while a force of magnitude F is applied to it at an angle θ as shown in figure. The coefficient of kinetic friction is μ_k. Then, the block's acceleration 'a' is given by: (g is acceleration due to gravity)



(1) $\frac{F}{m}\cos\theta+\mu_k\left(g-\frac{F}{m}\sin\theta\right)$
(2) $\frac{F}{m}\cos\theta+\mu_k\left(g+\frac{F}{m}\sin\theta\right)$
(3) $\frac{F}{m}\sin\theta+\mu_k\left(g-\frac{F}{m}\cos\theta\right)$
(4) $\frac{F}{m}\sin\theta-\mu_k\left(g-\frac{F}{m}\cos\theta\right)$

6. In a Young's double slit experiment two slits are separated by 1 mm and the screen is placed one meter away. When a light of wavelength 600nm used, the fringe separation will be:

(1) 3 mm
(2) 0.30 mm
(3) 0.60 mm
(4) 6 mm

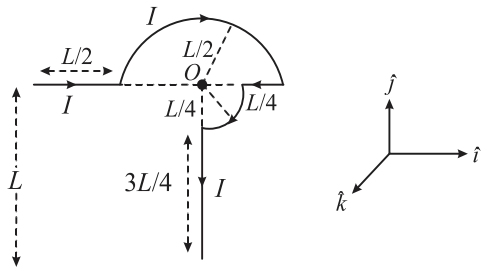
7. A ball is spun with angular acceleration $\alpha=3t^3-t^2+2t$ where t is in second and α is in rads^{-2}. At t = 0, the ball has angular velocity of 10 rads^{-1} and angular position of 4 rad. The most appropriate expression for the angular position of the ball at any time t is:

(1) $\frac{3}{20}t^5-\frac{t^4}{12}+\frac{t^3}{4}+10$
(2) $\frac{3}{20}t^5-\frac{t^4}{12}+\frac{t^3}{3}+10t+4$
(3) $\frac{3}{5}t^5-\frac{t^4}{12}+\frac{t^3}{3}+4$
(4) None of these

8. Consider two uniform discs of the same thickness and different radii $R_1 = R$ and $R_2=\frac{\alpha R}{2}$ made of the same material. If the ratio of their moment of inertia I_1 and I_2, respectively, about their axes is $I_1 : I_2 = 1 : 16$ then the value of α is:

(1) 16
(2) 8
(3) 4
(4) 2

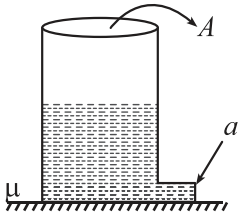
9. Which one of the following options represents the magnetic field $\vec{B}$ at *O* due to the current flowing in the given wire segments lying on the *xy* plane?



(1) $\frac{3}{2}\frac{\mu_0 I}{L}\hat{k}$ (2) $-\frac{3}{2}\frac{\mu_0 I}{L}\hat{k}$

(3) $-\frac{\mu_0 I}{2L}\hat{k}$ (4) $-\frac{\mu_0 I}{L}\hat{k}$

10. A light cylindrical vessel is kept on a horizontal surface. Area of base is *A*. A hole of cross-sectional area *a* is made just at its bottom side. If the coefficient of friction is μ then the acceleration of the vessel due to the impact force of the emerging liquid is ($a << A$):



(1) $g\left(\frac{2a}{A}-\mu\right)$ (2) $g\left(\frac{a}{A}-\mu\right)$

(3) $g\left(\frac{a}{A}+\mu\right)$ (4) None of these

11. The number density of molecules of a gas depends on their distance *r* from the origin as, $m(r)=n_0 e^{-\alpha r^3}$. Then the total number of molecules is proportional to:

(1) $n_0\alpha$

(2) $n_0\alpha^{-\frac{1}{2}}$

(3) $n_0\alpha^{-1}$

(4) $n_0^{1/3}\alpha$

12. From the photoelectric effect experiment, following observations are made. Identify which of these are correct

A. The stopping potential depends on both frequency and the work function of the metal.

B. The saturation current increases as the intensity of incident light increases.

C. The maximum kinetic energy of a photo electron depends on the frequency of the incident light.

D. Photoelectric effect can be explained by both particle and wave theory of light.

Choose the correct answer from the options given below:

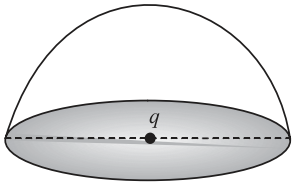
(1) A, B, C only

(2) A, C, D only

(3) B, D only

(4) B, C, D only

13. If a charge *q* is placed at the center of a closed hemi-spherical non-conducting surface, the total flux passing through the flat surface would be



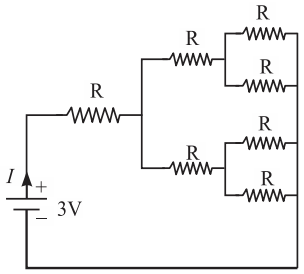
(1) $\frac{q}{\epsilon_0}$

(2) $\frac{q}{2\epsilon_0}$

(3) $\frac{q}{4\epsilon_0}$

(4) Zero

14. All resistance in figure are 1 Ω each. The value of current '*I*' is $\frac{a}{7}$A . The value of *a* is_____



(1) 8 (2) 10

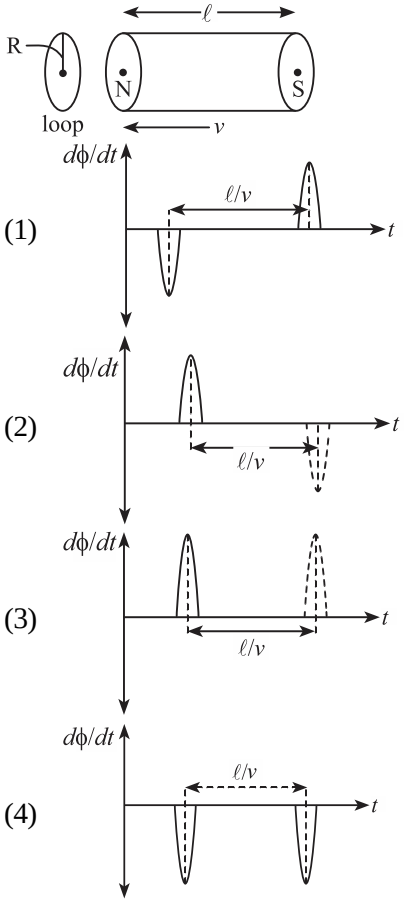
(3) 12 (4) 4

15. A rod with circular cross-section area 4 cm^2 and length 80 cm is wound uniformly with 400 turns of an insulated wire. If a current of 0.2 A flows in the wire windings, the total magnetic flux produced inside windings is $4\pi \times 10^{-6}$ Wb. The relative permeability of the rod is (Given: Permeability of vacuum $\mu_0 = 4\pi \times 10^{-7} NA^{-2}$)

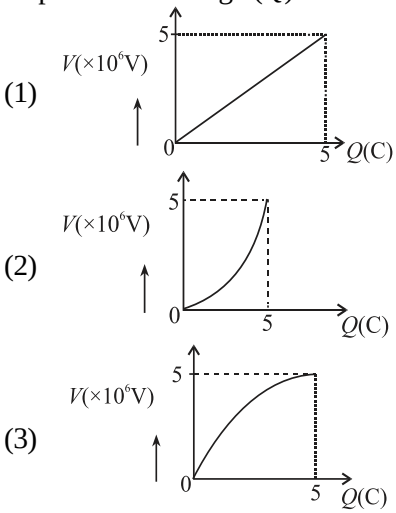
(1) 12.5 (2) 0.625

(3) 125 (4) 0.250

16. A bar magnet is passing through a conducting loop of radius R with velocity v. The radius of the bar magnet is such that it just passes through the loop. The rate of change of magnetic flux in the loop can be represented by the approximate curve:



17. A condenser of 1μF capacitance is charged steadily from 0 to 5 C. Which of the following graph represents correctly the variation of potential difference (V) across its plates with respect to the charge (Q) on the condenser?



(4) None of these

18. An EM wave propagating in x-direction has a wavelength of 4 mm. The electric field vibrating y-direction has maximum magnitude of 30 Vm^{-1}. Choose the correct equations for electric and magnetic field if the EM wave if propagating in vacuum

(1) $E_y = 30\sin\left[\frac{\pi}{2}\times10^3\left(x-3\times10^8t\right)\right]\hat{j}\ Vm^{-1}$

$B_z = 2\sin\left[\frac{\pi}{2}\times10^3\left(x-3\times10^8t\right)\right]\hat{k}\,T$

(2) $E_y = 30\sin\left[\frac{\pi}{2}\times10^3\left(x-3\times10^8t\right)\right]\hat{j}\ Vm^{-1}$

$B_z = 10^{-7}\sin\left[\frac{\pi}{2}\times10^3\left(x-3\times10^8t\right)\right]\hat{k}\ T$

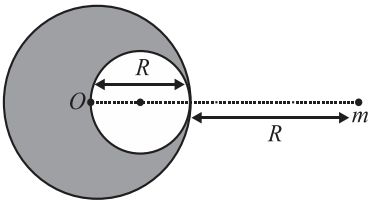
(3) $E_y = 2\times10^{-7}\sin\left[\frac{\pi}{2}\times10^3\left(x-3\times10^8t\right)\right]\hat{j}\,Vm^{-1}$

$B_z = 30\sin\left[\frac{\pi}{2}\times10^3\left(x-3\times10^8t\right)\right]\hat{k}T$

(4) $E_y = 10^{-7}\sin\left[\frac{\pi}{2}\times10^4\left(x-6\times10^8t\right)\right]\hat{j}\,Vm^{-1}$

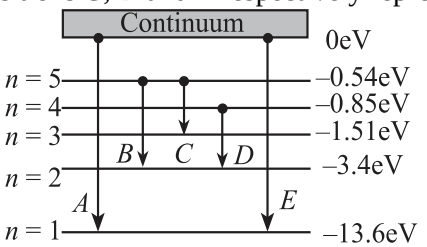
$B_z = 30\sin\left[\frac{\pi}{2}\times10^4\left(x-6\times10^8t\right)\right]\hat{k}\,T$

19. A solid sphere of radius R and mass M gravitationally attracts a particle of mass m placed at $2R$ from its centre with a force F_1. Now a spherical cavity of radius $\left(\frac{R}{2}\right)$ is made in the sphere (as shown in figure) and the force becomes F_2. The value of F_1:F_2 is



(1) 9 : 8 (2) 9 : 7
(3) 8 : 9 (4) 7 : 9

20. In the given, figure, the energy levels of hydrogen atom have been shown along with some transitions marked A, B, C, D and E. The transitions C, D and E respectively represent:

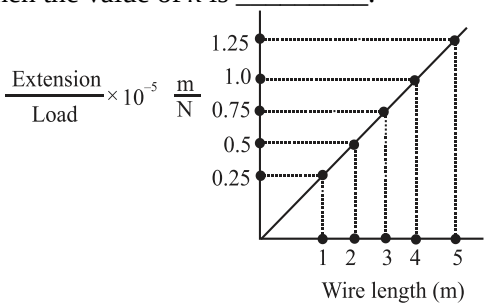


(1) The 3rd member of Balmer series, 2nd member of Lyman series and the series limit of Lyman series
(2) The 3rd member of Paschen series, 2nd member of Balmer series and the series limit of Balmer series
(3) The ionization potential of hydrogen, 3rd member of Paschen series and 2nd member of Balmer series
(4) The 2rd member of Paschen series, 2nd member of Balmer series and the series limit of Lyman series

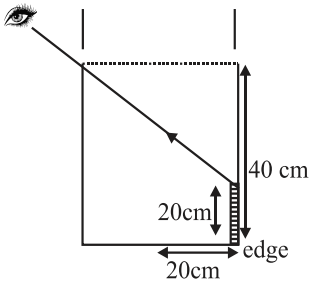
Integer Type Questions

21. Three harmonic waves having equal frequency v and same intensity I_0, have phase angles $0, \frac{\pi}{4}$ and $-\frac{\pi}{2}$ respectively. When they are superimposed the intensity of the resultant wave is close to xI_0. Find x.

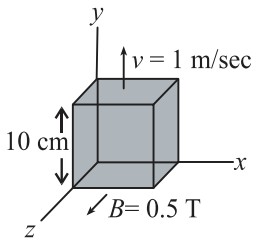
22. In an experiment to determine the Young's modulus, steel wires of five different lengths (1, 2, 3, 4, and 5m) but of same cross section (1 mm^2) were taken and curves between extension and load were obtained. The slope (extension/load) of the curves were plotted with the wire length and the following graph is obtained. If the Young's modulus of given steel wires is $x \times 10^{11}$ Nm^{-2}, then the value of x is _____.



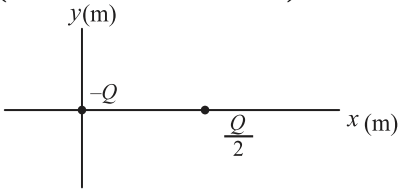
23. An observer can see through a small hole on the side of a jar (radius 20 cm) at a point at height of 20 cm from the bottom (see figure). The hole is at a height of 40 cm. When the jar is filled with a liquid upto a height of 30 cm the same observer can see the edge at the bottom of the jar. If the refractive index of the liquid is $\sqrt{\frac{x}{5}}$, where x is an integer, the value of x is _____.



24. A metallic cube of side 10 cm moving along y-axis at a uniform velocity of 1 ms^{-1}. In a region of uniform magnetic field of magnitude 0.5 T directed along z-axis. In equilibrium the magnitude of the potential difference between the faces of higher and lower potential developed because of the motion through the field will be _____mV.



25. Two point charges $-Q$ and $+Q/2$ are placed in the xy-plane at the origin (0, 0) and a point (2, 0) respectively, as shown in the figure. This results in an equipotential circle of radius R and potential $V = 0$ in the xy-plane. All lengths are measured in meters. (All distances in SI units)



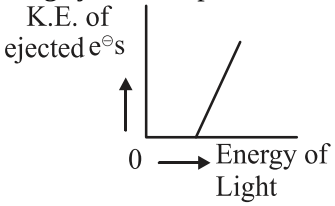
The value of R is $\frac{p}{3}$ meter. Find p.

SECTION-II (CHEMISTRY)

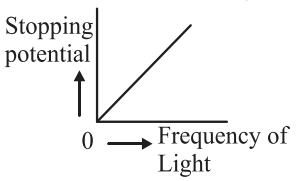
Single Correct Type Questions

26. A radiation with energy greater than threshold energy is incident on the surface of a metal. Which one of the following graphs is incorrect regarding ejection of photoelectrons?

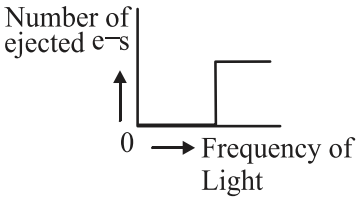
(1)



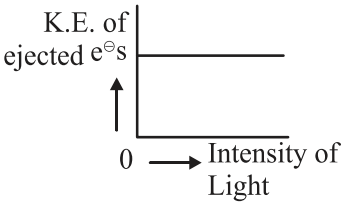
(2)



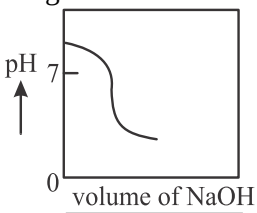
(3)

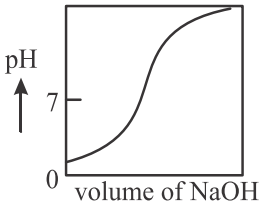


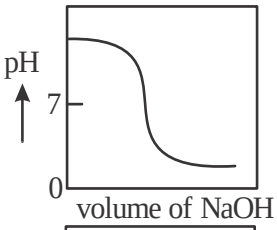
(4)

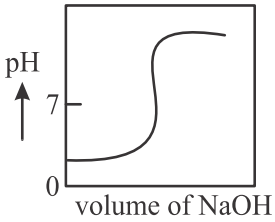


27. 0.1 M HCl is taken in a titration flask and titrated against 0.1 M NaOH solution. Which one of the following titration curves is correct?

(1) 

(2) 

(3) 

(4) 

28. Identify the compound 'P' from in the following sequence of reactions.

(P) $\xrightarrow[\text{(ii) Conc. } H_2SO_4/\Delta]{\text{(i) MeMgBr/}H_2O}$ (Q)

(Q) $\xrightarrow[\text{(ii) Zn/}H_2O]{\text{(i) } O_3}$ R + S

(R) $\xrightarrow[\text{(ii) }\Delta]{\text{(i) Conc. NaOH}}$

$PhCOO^-Na^+$ $PhCH_2OH$

(S) $\xrightarrow[\Delta]{Ba(OH)_2} H_3C-\overset{CH_3}{\overset{|}{C}}=CH-\overset{O}{\overset{||}{C}}-CH_3$

(1) $Ph-CH=CHMe$

(2) $Ph-CH_2-\overset{O}{\overset{||}{C}}-Me$

(3) $Ph-CH=C(Me)_2$

(4) $Ph-\overset{O}{\overset{||}{C}}-CH(Me)_2$

29. $\underset{\text{(Salt)}}{M^{2+}} \xrightarrow[OH^-]{\text{Reagent 'R'}}$ Bright red ppt. $\underset{\text{(square planar complex)}}{}$

Group of basic radicals to which M^{2+} ion belong and number of five membered chelate rings in the complex formed are, respectively,

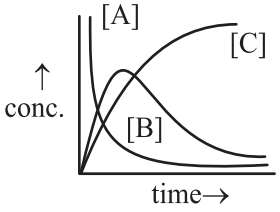
(1) III, 4
(2) IV, 4
(3) IV, 2
(4) V, 2

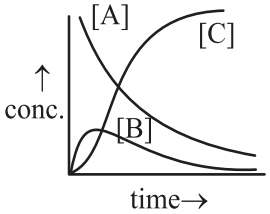
30. Which of the following simplest processes individually contribute to the standard electrode potential $\left(E^o_{M^+/M}\right)$ in aqueous solution?

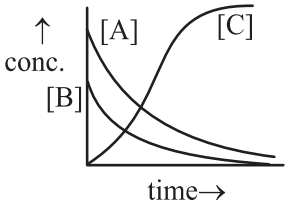
(A) $M(s) \rightarrow M^+(g) + e^-$
(B) $M(s) \rightarrow M(g)$
(C) $M(g) \rightarrow M^+(g)+e^-$
(D) $M^+(g) \xrightarrow{H_2O} M^+(aq.)$

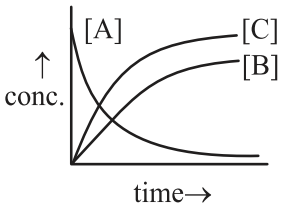
(1) A, B and C only
(2) B, C and D only
(3) A, B and D only
(4) B and C only

31. For the consecutive reaction, $A \xrightarrow{k_1} B \xrightarrow{k_2} C$, identify the correct graph if $k_2 >> k_1$

(1) 

(2) 

(3) 

(4) 

32. Consider the following statements:

Statement-I: $FeCl_2$ has greater covalent character than $ZnCl_2$

Statement-II: Fe^{2+} is less polarising than Zn^{2+}

In the light of the above statements, choose the most appropriate answer from the options given below

(1) Both Statement (I) and Statement (II) are correct.
(2) Both Statement (I) and Statement (II) are incorrect.
(3) Statement (I) is correct but Statement (II) is incorrect.
(4) Statement (I) is incorrect but Statement (II) is correct.

33. Which one of the following 3d series elements, represented by atomic number, has $E^{o}_{M^{3+}/M^{2+}} < 0$?

(1) Z = 25 (2) Z = 27
(3) Z = 24 (4) Z = 26

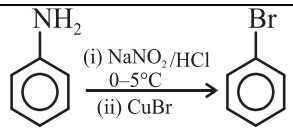
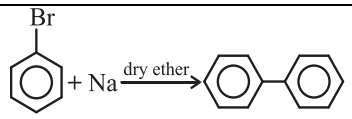
34. Match Column-I with Column-II

Column-I (Ions/molecules)		Column-II (Hybridisation)	
A	$[NiCl_4]^{2-}$	P	sp^3d
B	$[Fe(CN)_6]^{4-}$	Q	d^2sp^3
C	PF_5	R	sp^3
D	$[CoF_6]^{3-}$	S	sp^3d^2

Select the correct answer using the codes given below:

	A	B	C	D
(1)	Q	S	P	R
(2)	R	Q	P	S
(3)	R	P	S	Q
(4)	Q	P	S	R

35. Match Column-I with Column-II

Column-I		Column-II	
A		P	Fittig reaction
B		Q	Sandmeyer Reaction
C	$EtCl + AgF \rightarrow Et–F + AgCl$	R	Finkelstein reaction
D	$MeBr + NaI \xrightarrow{\text{acetone}} MeI + NaBr$	S	Swarts Reaction

Select the correct answer using the codes given below

	A	B	C	D
(1)	Q	P	S	R
(2)	R	Q	S	P
(3)	S	Q	R	P
(4)	Q	P	R	S

36. An element with Z = 13 forms an amphoteric oxide. This oxide reacts with both HCl as well as NaOH. With 1 mol of oxide, ratio of moles of HCl & NaOH required for complete reaction, is

(1) 3 : 1 (2) 1 : 3
(3) 1 : 1 (4) 1 : 2

37. Consider the following statements

Statement I: In aqueous solution, bare proton does not exist.

Statement II: Proton forms species like H_3O^+, $H_5O_2^+, H_7O_3^+$ in aqueous solution

In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both Statement (I) and Statement (II) are correct.
(2) Statement (I) is correct but Statement (II) is incorrect
(3) Statement (I) is incorrect but Statement (II) is correct
(4) Both Statement (I) and Statement (II) are incorrect.

38. Which one of the following symbols represents a hypothetical element having Z = 119?

(1) Uun (2) Nne
(3) Uue (4) Une

39. Identify the correct increasing order of bond angle T–C–T, where C denotes central atom and T denotes terminal atom

(1) $NO_2 < NO_2^- < NO_2^+$
(2) $F_2O < Cl_2O < H_2O$
(3) $ClO_2^- < OCl_2 < ClO_2$
(4) $NH_3 < PH_3 < AsH_3$

40. From the reactions given below, non-metal displacement (redox) reactions are

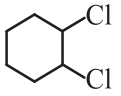
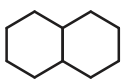
(A) $2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$
(B) $TiCl_4(l) + 2Mg(s) \xrightarrow{\Delta} Ti(s) + 2MgCl_2(s)$
(C) $2H_2O(l) + 2F_2(g) \rightarrow 4HF(aq) + O_2(g)$
(D) $2NaH(s) \xrightarrow{\Delta} 2Na(s) + H_2(g)$

(1) A and B only (2) B and C only
(3) A and C only (4) A and D only

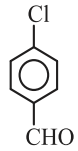
41. Given below are few characteristics of group 13 elements. The incorrect statement is

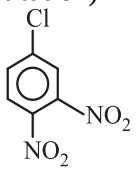
(1) Crystalline boron is unreactive
(2) Aluminium and gallium react with both acids & alkalies
(3) Indium forms basic oxide
(4) Tl^{3+} is highly stable in aqueous solution

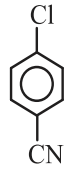
42. Which one of the following compounds does not exist in diastereomeric forms?

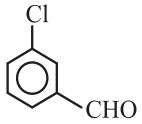
(1) Ph – N = N – Ph
(2) $(CH_3)_2C = CH(CH_3)$
(3) 
(4) 

43. Which one of the following halides is least reactive towards EAS but most reactive towards NAS?
(EAS: Electrophilic aromatic substitution, NAS : Nucleophilic aromatic substitution)

(1) 

(2) 

(3) 

(4) 

44. Given below are two statements, one is labelled as Assertion(A) and the other is labelled as Reason(R)
Assertion (A): At 298 K, molality of an aqueous solution of sucrose is 2.5 m. On heating the solution to 350 K, molality of solution remains unaffected
Reason (R): Molality is independent of temperature as mass does not change with temperature
In the light of the above statements, choose the most appropriate answer from the options given below
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

45. Consider the following reactions
(i) $\underset{(\text{excess})}{\text{MeBr}} + NH_3 \xrightarrow{\Delta}$
(ii) $\text{MeBr} + \underset{(\text{excess})}{NH_3} \xrightarrow{\Delta}$
Which one of the following statements is correct regarding the products of above reactions?
(1) Both reactions (i) & (ii) give exclusively primary amine as the major product
(2) Reaction (i) gives a mixture of 1°, 2°, 3° amine and quaternary ammonium salt while reaction (ii) gives 1° amine as major product
(3) Both reactions give a mixture of 1°, 2° & 3° amines as products.
(4) Reaction (i) gives majorly 1° amine while reaction (ii) gives a mixture of 1°, 2° & 3° amines as products.

Integer Type Questions

46. How many of the given compounds will give a purple colouration with $CuSO_4$(aq) + NaOH + Rochelle salt? (Ignore any faint colouration)
Arginine, Glycylalanine, Urea, Glutathione, Glycylhistidyllysine

47. 'V' L of a perfect gas is expanded isothermally against a constant external pressure of 2 atm till final volume becomes 5V. the amount of heat absorbed in this process is 20 kJ. The value of 2V is_____. (1 atm L = 100 J)

48. Consider the following electrochemical cell:
Pt(s) $|H_2(g, 1\text{ bar})|$ HCl(aq)|AgCl(s)|Ag(s)
When HCl solution of pH = 5 is used then the cell potential is found to be 0.80 V. The cell potential when HCl solution of pH = 6 is used, is 'x' volt. The value of 100 x is_____.
$\left\{\text{Given: } \frac{2.303RT}{F} = 0.06\,\text{V at } 298\,\text{K}\right\}$

49. At 300 K, gases XY_2 and PQ_2 decomposes independently following I order kinetics. If the time taken for 90% decomposition of XY_2(g) and PQ_2(g) is 12 min and 9 min respectively, then the value of $\frac{3\times\left(t_{\frac{1}{2}}\right)_{XY_2}}{\left(t_{\frac{1}{2}}\right)_{PQ_2}}$ is _____.[$\log_{10}2 = 0.30$]

50. Aqua complex of Co^{2+} i.e. $[Co(H_2O)_6]^{2+}$ is pink in colour while chloro complex i.e. $[CoCl_4]^{2-}$ is blue in colour. The value of $\frac{2\times\left|(\text{CFSE})_{\text{aqua}}\right|}{\left|(\text{CFSE})_{\text{Chloro}}\right|}$ is_____.
(Consider splitting effect of chloride and aqua ligands as same)

SECTION-III (MATHEMATICS)

Single Correct Type Questions

51. The number of integral values of 'k' for which the inequality $x^2 - 2(3k-3)x + 8k^2 - 63 > 0$ is valid for every $x \in R$ is:
(1) 5 (2) 2
(3) 4 (4) 6

52. Four dice are rolled. If the probability of getting different numbers on the four dice is $\frac{p}{q}$, where p and q are coprime, then $q - p$ is equal to
(1) 4 (2) 13
(3) 1 (4) 14

53. The equation $e^{4x} + 8e^{3x} + 10e^{2x} - 8e^x + 1 = 0$, $x \in R$ has:
(1) Two solutions and both are negative
(2) No solution
(3) Four solutions two of which are negative
(4) Two solutions and only one of them is negative

54. If sum of the first 12 terms of the series $\log_{\frac{1}{7^2}} x + \log_{\frac{1}{7^3}} x + \log_{\frac{1}{7^4}} x + \ldots\ldots$, where $x > 0$ is 270, then x is equal to

(1) 27 (2) 49
(3) 343 (4) 81

55. Let $A = \begin{pmatrix} 4 & -1 \\ 12 & -3 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & -6 \\ 1 & -2 \end{pmatrix}$. Then the number of elements in the set $\{(n, m)$: $n, m \in \{1, 2, \ldots\ldots, 10\}$ and $nA^n + mB^m = I\}$

(1) 0 (2) 2
(3) 3 (4) 4

56. If $\frac{5 + i\sin\theta}{2 - i\cos\theta}, \theta \in [0, 2\pi]$ is a real number, then an argument of $\sin\theta + i\cos\theta$ is:

(1) $\pi - \tan^{-1}\left(\frac{3}{4}\right)$ (2) $\tan^{-1}\left(\frac{5}{2}\right)$
(3) $\pi - \tan^{-1}\left(\frac{2}{5}\right)$ (4) $-\tan^{-1}\left(\frac{4}{3}\right)$

57. The number of ways of giving 24 identical oranges to 3 children such that each child gets atleast two oranges is_____

(1) 160 (2) 150
(3) 190 (4) 180

58. If the domain of the function $f(x) = \log_e(4x^2 + 15x + 9) + \cos^{-1}(4x + 3) + \sin^{-1}\left(\frac{10x + 6}{3}\right)$ is $(\alpha, \beta]$, then $48|\alpha - \beta|$ is equal to

(1) 63 (2) 12
(3) 72 (4) 54

59. If $({}^{40}C_1)^2 + 2({}^{40}C_2)^2 + 3({}^{40}C_3)^2 + \ldots + 40({}^{40}C_{40})^2 = \frac{\alpha \cdot 80!}{(40!)^2}$, then α is equal to

(1) 30 (2) 60
(3) 20 (4) 10

60. Let α, β be the roots of the quadratic equation $x^2 + \sqrt{8}x + 4 = 0$. Then $\frac{\alpha^{25} + \beta^{25} + \alpha^{14} + \beta^{14}}{\alpha^{17} + \beta^{17} + \alpha^{10} + \beta^{10}}$ is equal to

(1) 729 (2) 72
(3) 256 (4) 9

61. Five number x_1, x_2, x_3, x_4, x_5 are randomly selected from the number 1, 2, 3,....., 18 and are arranged in the increasing order ($x_1 < x_2 < x_3 < x_4 < x_5$). The probability that $x_2 = 9$ and $x_4 = 12$ is

(1) $\frac{1}{136}$ (2) $\frac{1}{72}$
(3) $\frac{4}{357}$ (4) $\frac{1}{34}$

62. Let the complex number $z = x + iy$ be such that $\frac{2z - 3}{2z + 1}$ is purely imaginary. If $x + y^2 = 0$, then $y^4 + 2y^2$ is equal to

(1) $\frac{3}{2}$ (2) $\frac{3}{4}$
(3) $\frac{2}{3}$ (4) $\frac{4}{3}$

63. Let $X = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ and $Y = \alpha I - \beta X + \gamma X^2$, where $\alpha, \beta, \gamma \in \mathbb{R}$. If $Y^{-1} = \begin{bmatrix} \frac{1}{5} & \frac{-2}{5} & \frac{1}{5} \\ 0 & \frac{1}{5} & \frac{-2}{5} \\ 0 & 0 & \frac{1}{5} \end{bmatrix}$, then $(\alpha + \beta - \gamma)^2$ is equal to____

(1) 75 (2) 400
(3) 125 (4) 150

64. If $\tan\alpha = 1$ and $\sec\beta = -\frac{5}{4}$, where $\pi < \alpha < \frac{3\pi}{2}$ and, $\frac{\pi}{2} < \beta < \pi$, then the value of $\tan(\alpha + \beta)$ and the quadrant in which $\alpha + \beta$ lies, respectively are

(1) $-\frac{1}{7}$ and IVth quadrant
(2) 7 and Ist quadrant
(3) –7 and IVth quadrant
(4) $\frac{1}{7}$ and Ist quadrant

65. If the inverse trigonometric functions take principal values, then $\sin^{-1}\left(\frac{3}{10}\sin\left(\cot^{-1}\left(\frac{4}{3}\right)\right) + \frac{2}{5}\cos\left(\cot^{-1}\left(\frac{4}{3}\right)\right)\right)$ is equal to

(1) 0 (2) $\frac{\pi}{4}$
(3) $\frac{\pi}{3}$ (4) $\frac{\pi}{6}$

66. Consider four boxes, each containing 10 balls labelled 1, 2, ..., 10. Suppose one ball is randomly drawn from each of the boxes. Denoted by n_i, the label of the ball drawn from the i_{th} box, (i = 1,2,3,4). Then, the number of ways in which the balls can be chosen such that $n_1 < n_2 < n_3 < n_4$ is:

(1) 210
(2) 82
(3) 240
(4) 164

67. $\lim_{n\to\infty}\left(1+\frac{1^2-2^2+3^2-4^2.....-n^2}{n^3}\right)^n$ is equal to

(1) e (2) 0

(3) $e^{-\frac{1}{2}}$ (4) $e^{\frac{1}{2}}$

68. Let $f(x)=\sum_{k=1}^{9}(2k-1)x^k, x\in R$

If $f'(2)+2f(2)=(2n+1)\times 2191$ then n is equal to

(1) 21 (2) 23

(3) 33 (4) 17

69. Let $I(x)=\int\frac{(x+1)}{x(1+xe^x)^4}dx, x>0$, then $I(x)$ is

(1) $\ln\left(\frac{xe^x}{xe^x+1}\right)+\frac{1}{\left(xe^x+1\right)}+\frac{1}{2\left(xe^x+1\right)^2}+\frac{1}{3\left(xe^x+1\right)^3}+C$

(2) $\ln\left(\frac{xe^x+1}{xe^x}\right)+\frac{1}{\left(xe^x+1\right)}-\frac{1}{2\left(xe^x+1\right)^2}+\frac{1}{3\left(xe^x+1\right)^3}+C$

(3) $\ln\left(\frac{xe^x+1}{xe^x}\right)-\frac{2}{\left(xe^x+1\right)}+\frac{1}{2\left(xe^x+1\right)^2}+\frac{1}{3\left(xe^x+1\right)^3}+C$

(4) $\ln\left(\frac{xe^x+1}{xe^x}\right)+\frac{1}{\left(xe^x+1\right)}+\frac{3}{2\left(xe^x+1\right)^2}-\frac{1}{3\left(xe^x+1\right)^3}+C$

70. Let the solution curve $y=y(x), 0<x<\frac{\pi}{2}$ of the differential equation $(\ln\cos x)^2\,dy = \tan x\,(3y\ln\cos x+1)\,dx$, $y\left(\frac{\pi}{4}\right)=\frac{1}{2\ln\sqrt{2}}$. If $2y\left(\frac{\pi}{6}\right)=\frac{1}{\ln m-\ln\sqrt{n}}$, where m and n are co-prime, then $m+n$ is equal to

(1) 5 (2) 6

(3) 7 (4) 12

Integer Type Questions

71. The area of the region enclosed by $y\le 16x^2$, $x^2\le 25y$ and $y\le 9$ is equal to

72. The value of $\int_0^{\pi/8}\left[\sec x+2\left[\tan x+3\left[\cos x+4\left[\sin x\right]\right]\right]\right]dx=\frac{\pi}{k}$ (where [.] denote greatest integer function), then value of k is

73. If the point $\left(\beta,\frac{5\sqrt{3}}{3}\right)$ lies on the curve traced by the mid-points of the line segments of the lines $x\cos\theta+y\sin\theta=5$ between the coordinates axes, then $|\beta|$ is equal to

74. Let $\vec{a}=5\hat{i}-\hat{j}-3\hat{k}$ and $\vec{b}=\hat{i}-2\hat{j}+2\hat{k}$ be two vectors. If projection length of $\vec{a}$ along $\vec{b}$ is k, then $3k$ is equal to

75. If the foot of the perpendicular drawn from (–1, 5, 5) to the line passing through the point (3, 2, 1) and perpendicular to vectors $\left(\hat{i}+2\hat{j}+\hat{k}\right)$ and $\left(-3\hat{j}+\hat{k}\right)$ is (α,β,γ) then $\alpha+\beta+\gamma$ is equal to

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